Data-Driven Modeling of MAPK Dynamics

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Collaborators & Acknowledgement

Collaborators

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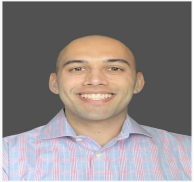

Liam O'Brien (OSU)

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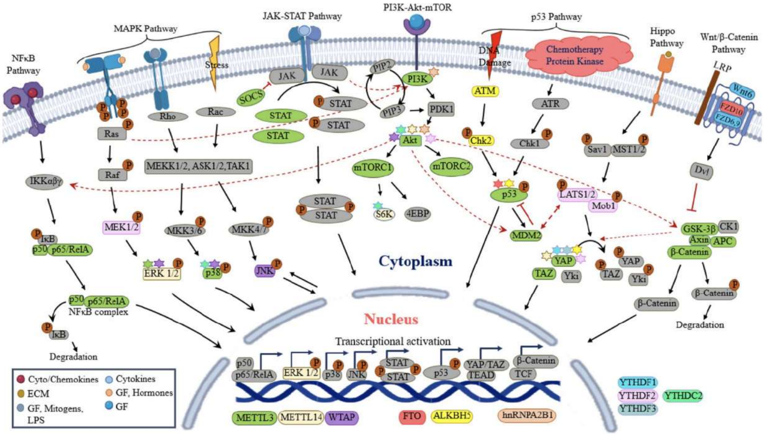
Outline

Introduction: Mitogen-Activated Protein Kinase (MAPK)

- Is a family of proteins thats play a crucial role in cellular signaling pathways
- They act as a command and control center for a wide range of essential cell functions such as.
    - Cell growth and division (proliferation)
    - Differentiation (specializing into specific cell types)
    - Apoptosis (cell death)
    - Stress responses (such as from UV radiation, osmotic shock, or inflammation)
    - Gene expression

Key MAPK Pathways

- **ERK Pathway (This Presentation's Focus):** Responds to growth factors. Controls cell proliferation, survival, and differentiation.
- **JNK Pathway:** Responds to cellular stress and cytokines. It controls cell apoptosis.
- **p38 Pathway:** Responds to UV irradiation and responsible for modulating inflammation.



MAPK Signaling Pathway via ERK

The Raf-MEK-ERK pathway represents the core MAPK signaling cascade, extensively studied due to its critical role in tumor growth and cancer progression.

The signal transduction cascade proceeds as follows:

MAPK Signaling Pathway via ERK

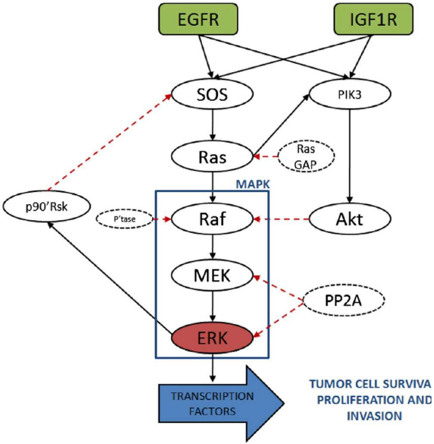
Ligand $\rightarrow$ **EGFR** $\rightarrow$ **Ras** $\rightarrow$ **Raf** $\rightarrow$ **MEK** $\rightarrow$ **ERK** $\rightarrow$ **Nuclear Targets**

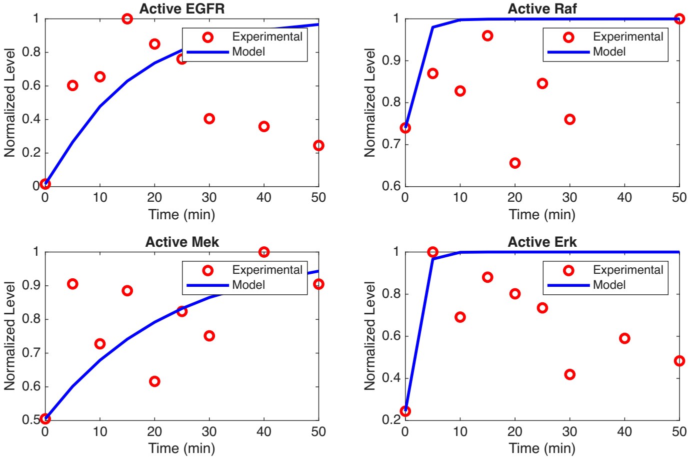
When this pathway is stuck in the activated mode, it can lead to diseases like cancer.

Description of the Biological Data

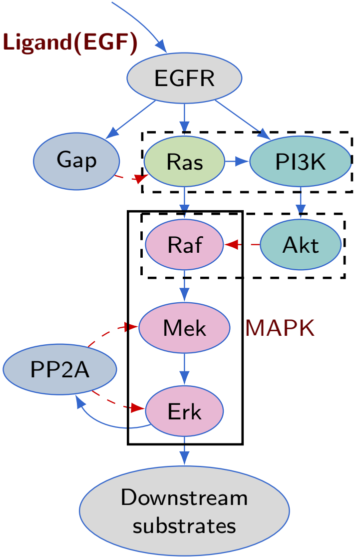
	A	B	C	D	E	J	M	AC	AD	AX	BI	DC	DD
1	**Sample ID**	**Cell line**	**Mutation**	**Treatment**	**Time**	**AKT_S473**	**EGFR_Y1068**	**MEK_1_2_S217_221**	**ERK_T202_Y204**	**PI3K_p85_p55_**	**ATP_Citrate_Lya**	**Raf_S259**	**Ras_GRF1_S916**
2	1	A549	EGFR wild-type KRAS mutant	Baseline	T0	3435.783248	360.3227235	8484.527817	6884.307969	11114.43491	5972.970745	14002.62276	9461.627931
3	2	A549	EGFR wild-type KRAS mutant	Baseline	T0	2643.872862	371.2960849	7990.43165	6298.080258	10519.64436	5115.344167	12258.80284	8544.135756
4	3	A549	EGFR wild-type KRAS mutant	Baseline	T0	3508.697296	304.2957028	11259.87131	8655.930611	11754.59775	6026.974804	13821.77321	9490.057896
5	4	A549	EGFR wild-type KRAS mutant	EGF	5 min	5889.932143	15244.89496	21482.63861	33860.36059	12394.40191	6008.919416	13480.5047	9623.852501
6	5	A549	EGFR wild-type KRAS mutant	EGF	5 min	4740.982863	13521.01045	14457.96096	29436.77323	8242.015548	8250.257154	12926.05095	10076.89453
7	6	A549	EGFR wild-type KRAS mutant	EGF	5 min	4959.200536	12052.1737	13780.36645	26108.07876	8928.468174	8946.341682	11170.14752	8760.428241
8	7	A549	EGFR wild-type KRAS mutant	EGF	10 min	3721.939276	14720.55539	10636.00596	21418.28172	10239.41948	9471.098121	11533.3681	8681.939645
9	8	A549	EGFR wild-type KRAS mutant	EGF	10 min	5658.984816	15290.70229	14214.25019	21375.49106	9348.766132	9956.693605	11591.18068	9339.427169
10	9	A549	EGFR wild-type KRAS mutant	EGF	10 min	6222.954426	14328.41195	15108.30913	18996.31395	11731.11374	12136.83452	11430.0355	8425.343766
11	10	A549	EGFR wild-type KRAS mutant	EGF	15 min	4251.382666	26108.07876	14559.52129	28282.54192	11114.43491	11544.91191	12606.91043	10530.17352
12	11	A549	EGFR wild-type KRAS mutant	EGF	15 min	5156.432976	19148.88286	17641.33272	24587.65229	13835.5943	13121.4077	12234.31151	10373.3961
13	12	A549	EGFR wild-type KRAS mutant	EGF	15 min	4934.465856	22471.44021	16415.81253	25848.29312	11837.16823	13725.35906	12394.40191	10732.16018
14	13	A549	EGFR wild-type KRAS mutant	EGF	20 min	5866.421172	22004.44144	11920.31873	25084.3673	10509.13575	10425.39416	9887.240422	7427.912234
15	14	A549	EGFR wild-type KRAS mutant	EGF	20 min	3310.98015	15048.00096	9509.055627	24587.65229	10905.25357	13095.18052	10026.63466	9237.25059
16	15	A549	EGFR wild-type KRAS mutant	EGF	20 min	5218.683362	20496.31617	12406.7957	22004.44144	10519.64436	10742.89181	9986.601602	8946.341682
17	16	A549	EGFR wild-type KRAS mutant	EGF	25 min	8769.196597	22697.26548	15521.78218	26635.48372	10311.34656	13780.36645	12369.63967	10066.8186
18	17	A549	EGFR wild-type KRAS mutant	EGF	25 min	6198.114122	16764.18822	16236.21784	23388.51619	14868.50284	14400.23506	12913.12614	11293.69381
19	18	A549	EGFR wild-type KRAS mutant	EGF	25 min	3916.680503	12028.09519	13480.5047	15677.78228	13891.04348	8586.956208	11920.31873	10894.34937
20	19	A549	EGFR wild-type KRAS mutant	EGF	30 min	2635.952445	7762.034672	9585.436758	10435.82899	13453.57259	8151.845034	10938.02129	9018.202835
21	20	A549	EGFR wild-type KRAS mutant	EGF	30 min	3854.513081	8991.186435	12100.47543	12040.12268	11932.24984	7065.647332	11418.61745	9163.652704
22	21	A549	EGFR wild-type KRAS mutant	EGF	30 min	6953.498005	10667.95448	19574.83261	14957.97552	14086.89836	5972.970745	12040.12268	9145.345039
23	22	A549	EGFR wild-type KRAS mutant	EGF	40 min	11192.50855	8127.431788	19535.7249	20150.81981	12406.7957	5854.700903	12173.29124	8813.144981
24	23	A549	EGFR wild-type KRAS mutant	EGF	40 min	6568.231925	7615.947822	15630.81535	14853.63576	12644.77917	6399.658695	12977.85494	9265.006354
25	24	A549	EGFR wild-type KRAS mutant	EGF	40 min	8629.999493	8527.057666	19751.7978	17747.50811	12926.05095	6444.613267	13507.49073	9966.649803
26	25	A549	EGFR wild-type KRAS mutant	EGF	50 min	8450.659942	6329.650241	15994.50181	15153.70604	11259.87131	6721.050363	13602.37446	10530.17352
27	26	A549	EGFR wild-type KRAS mutant	EGF	50 min	6342.32129	4846.44178	14868.50284	14400.23506	13121.4077	8433.776734	13480.5047	10239.41948
28	27	A549	EGFR wild-type KRAS mutant	EGF	50 min	10188.349	5426.230522	18826.11747	13643.24639	13413.26887	8716.734425	13615.98914	11014.85572
29													

Existing Model from Bianconi et al. 2012





PROPOSED MODEL 1



The modeling kinetics: Examples

$$\text{Ras}^* + \text{PI3K} \xrightarrow{k_i} \text{Ras}^* + \text{PI3K}^*$$

$$\text{Akt}^* + \text{Raf}^* \xrightarrow{k_i} \text{Akt}^* + \text{Raf}$$

These kinetics are Michaelis–Menten kinetics.

New Models of MAPK Pathway

The series of phosporylation and dephosphorylation meachanisms are presented as;

$$\text{Ligand} + \text{EGFR} \xrightarrow{k_1} \text{Ligand} + \text{EGFR}^*$$
$$\text{EGFR}^* + \text{PI3K} \xrightarrow{k_2} \text{EGFR}^* + \text{PI3K}^*$$
$$\text{Ras}^* + \text{PI3K} \xrightarrow{k_3} \text{Ras}^* + \text{PI3K}^*$$
$$\text{PI3K}^* + \text{Akt} \xrightarrow{k_4} \text{PI3K}^* + \text{Akt}^*$$
$$\text{EGFR}^* + \text{Ras} \xrightarrow{k_5} \text{EGFR}^* + \text{Ras}^*$$
$$\text{Gap}^* + \text{Ras}^* \xrightarrow{k_6} \text{Gap}^* + \text{Ras}$$
$$\text{Akt}^* + \text{Raf}^* \xrightarrow{k_7} \text{Akt}^* + \text{Raf}$$
$$\text{Ras}^* + \text{Raf} \xrightarrow{k_8} \text{Ras}^* + \text{Raf}^*$$
$$\text{Raf}^* + \text{Mek} \xrightarrow{k_9} \text{Raf}^* + \text{Mek}^*$$
$$\text{Mek}^* + \text{Erk} \xrightarrow{k_{10}} \text{Mek}^* + \text{Erk}^*$$
$$\text{PP2A} + \text{Mek}^* \xrightarrow{k_{11}} \text{PP2A} + \text{Mek}^*$$
$$\text{PP2A} + \text{Erk}^* \xrightarrow{k_{12}} \text{PP2A} + \text{Erk}^*$$
$$\text{EGFR}^* + \text{Gap} \xrightarrow{k_{13}} \text{EGFR}^* + \text{Gap}^*$$
$$\text{Erk}^* + \text{PP2A} \xrightarrow{k_{14}} \text{Erk}^* + \text{PP2A}^*$$

New Models of MAPK Pathway

We define 19 state variables:

$$
\begin{gathered}
E^* = [\text{EGFR}^*], \quad R^* = [\text{Ras}^*], \quad F^* = [\text{Raf}^*], \quad M^* = [\text{Mek}^*], \quad K^* = [\text{Erk}^*] \\
A^* = [\text{Akt}^*], \quad P^* = [\text{PI3K}^*], \quad G^* = [\text{Gap}^*], \quad S = [\text{PP2A}^*] \\
L = [\text{Ligand}], \quad E = [\text{EGFR}], \quad R = [\text{Ras}], \quad F = [\text{Raf}], \quad M = [\text{Mek}], \quad K = [\text{Erk}] \\
A = [\text{Akt}], \quad P = [\text{PI3K}], \quad G = [\text{Gap Ras}], \quad S = [\text{PP2A}],
\end{gathered}
$$

The conservation Laws

$$
\begin{aligned}
&E_{\text{tot}} = E + E^*, && P_{\text{tot}} = P + P^*, && A_{\text{tot}} = A + A^*, && R_{\text{tot}} = R + R^*, \\
&F_{\text{tot}} = F + F^*, && M_{\text{tot}} = M + M^*, && K_{\text{tot}} = K + K^*, && G_{\text{tot}} = G + G^* \\
&S_{\text{tot}} = S + S^*
\end{aligned}
$$

9 total conservation laws reduce the system to 10 variables.

System of ODEs for Model 1

$$\dot{L} = -k_l L$$

$$\dot{E}^* = \frac{k_1 L(E_T - E^*)}{K_1 + (E_T - E^*)} - k_e E^*$$

$$\dot{P}^* = \frac{k_2 E^*(P_T - P^*)}{K_2 + (P_T - P^*)} + \frac{k_3 R^*(P_T - P^*)}{K_3 + (P_T - P^*)} - k_p P^*$$

$$\dot{A}^* = \frac{k_4 P^*(A_T - A^*)}{K_4 + (A_T - A^*)} - k_a A^*$$

$$\dot{R}^* = \frac{k_5 E^*(R_T - R^*)}{K_5 + (R_T - R^*)} - \frac{k_6 G^* R^*}{K_6 + R^*} - k_r R^*$$

$$\dot{F}^* = \frac{k_7 R^*(F_T - F^*)}{K_7 + (F_T - F^*)} - \frac{k_8 A^* F^*}{K_8 + F^*} - k_f F^*$$

$$\dot{M}^* = \frac{k_9 F^*(M_T - M^*)}{K_9 + (M_T - M^*)} - \frac{k_{10} S^* M^*}{K_{10} + M^*} - k_m M^*$$

$$\dot{K}^* = \frac{k_{11} M^*(K_T - K^*)}{K_{11} + (K_T - K^*)} - \frac{k_{12} S^* K^*}{K_{12} + K^*} - k_d K^*$$

$$\dot{G}^* = \frac{k_{13} E^*(G_T - G^*)}{K_{13} + (G_T - G^*)} - k_g G^*$$

$$\dot{S}^* = \frac{k_{14} K^*(S_T - S^*)}{K_{14} + (S_T - S^*)} - k_s S^*$$

Linearization of the system ODEs for Model 1

The linearized system is given by;

$$\dot{\vec{X}} = J_X \vec{X}$$

This system has only one equilibrium point which is:

$$X^* = (0, 0, 0, 0, 0, 0, 0, 0, 0, 0)$$

Theorem (Linear Stability of the Trivial Equilibrium)

The unique equilibrium point $X^* = \vec{0}$ of the linearized system is globally asymptotically stable.

Proof

The Jacobian matrix $J(X^*)$:

$$J(X^*) = \begin{pmatrix} -k_l & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ a_{21} & -k_e & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{32} & -k_p & 0 & a_{35} & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & a_{43} & -k_a & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & a_{52} & 0 & 0 & -k_r & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & a_{65} & -k_f & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & a_{76} & -k_m & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & a_{87} & -k_d & 0 & 0 \\ 0 & a_{92} & 0 & 0 & 0 & 0 & 0 & 0 & -k_g & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & a_{10,8} & 0 & -k_s \end{pmatrix}$$

The eigenvalues λ_i

$$-k_l, \ -k_e, \ -k_p, \ -k_a, \ -k_r, \ -k_f, \ -k_m, \ -k_d, \ -k_g, \ -k_s.$$

System of ODEs for Model 1

$$\dot{L} = -k_l L$$

$$\dot{E}^* = \frac{k_1 L(E_T - E^*)}{K_1 + (E_T - E^*)} - k_e E^*$$

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$$\dot{A}^* = \frac{k_4 P^*(A_T - A^*)}{K_4 + (A_T - A^*)} - k_a A^*$$

$$\dot{R}^* = \frac{k_5 E^*(R_T - R^*)}{K_5 + (R_T - R^*)} - \frac{k_6 G^* R^*}{K_6 + R^*} - k_r R^*$$

$$\dot{F}^* = \frac{k_7 R^*(F_T - F^*)}{K_7 + (F_T - F^*)} - \frac{k_8 A^* F^*}{K_8 + F^*} - k_f F^*$$

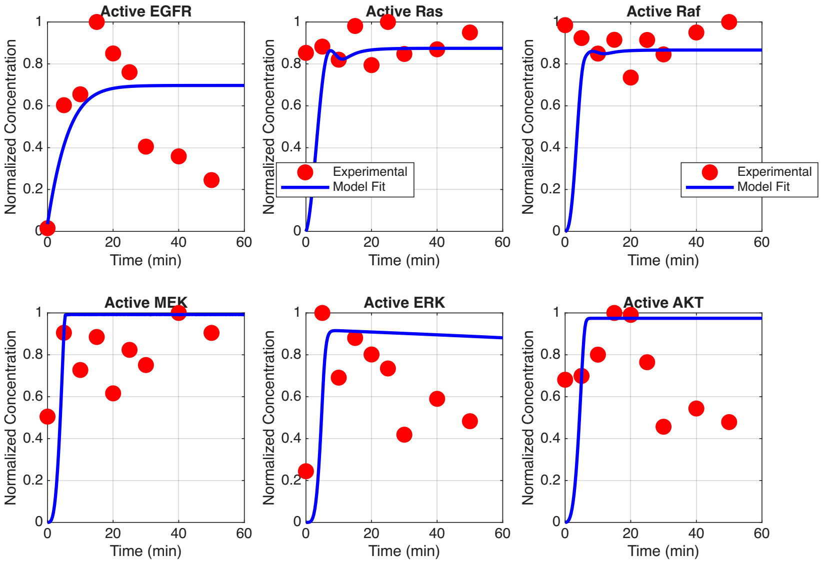
$$\dot{M}^* = \frac{k_9 F^*(M_T - M^*)}{K_9 + (M_T - M^*)} - \frac{k_{10} S^* M^*}{K_{10} + M^*} - k_m M^*$$

$$\dot{K}^* = \frac{k_{11} M^*(K_T - K^*)}{K_{11} + (K_T - K^*)} - \frac{k_{12} S^* K^*}{K_{12} + K^*} - k_d K^*$$

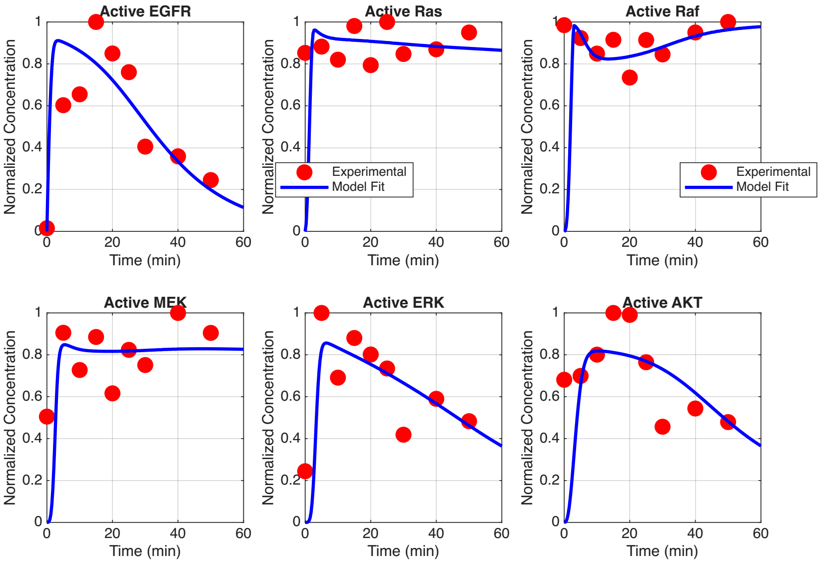
$$\dot{G}^* = \frac{k_{13} E^*(G_T - G^*)}{K_{13} + (G_T - G^*)} - k_g G^*$$

$$\dot{S}^* = \frac{k_{14} K^*(S_T - S^*)}{K_{14} + (S_T - S^*)} - k_s S^*$$

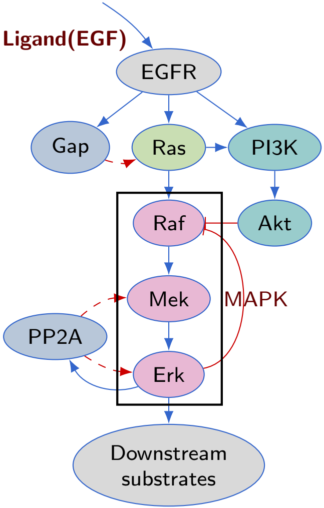
Validation of the System of ODEs Assuming L as a Constant

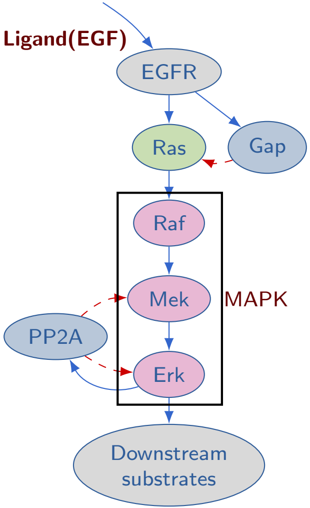


Validation of the System of ODEs using $\dot{L} = -k_l L$



Proposed Model 2





System of ODEs for Model 2

$$\dot{L} = -k_l L$$

$$\dot{E}^* = \frac{k_1 L(E_T - E^*)}{K_1 + (E_T - E^*)} - k_e E^*$$

$$\dot{P}^* = \frac{k_2 E^*(P_T - P^*)}{K_2 + (P_T - P^*)} + \frac{k_3 R^*(P_T - P^*)}{K_3 + (P_T - P^*)} - k_p P^*$$

$$\dot{A}^* = \frac{k_4 P^*(A_T - A^*)}{K_4 + (A_T - A^*)} - k_a A^*$$

$$\dot{R}^* = \frac{k_5 E^*(R_T - R^*)}{K_5 + (R_T - R^*)} - \frac{k_6 G^* R^*}{K_6 + R^*} - k_r R^*$$

$$\dot{F}^* = \frac{k_7 R^*(F_T - F^*)}{K_7 + (F_T - F^*)} - \frac{k_8 A^* F^*}{K_8 + F^*} - k_f F^*$$

$$\dot{M}^* = \frac{k_9 F^*(M_T - M^*)}{K_9 + (M_T - M^*)} - \frac{k_{10} S^* M^*}{K_{10} + M^*} - k_m M^*$$

$$\dot{K}^* = \frac{k_{11} M^*(K_T - K^*)}{K_{11} + (K_T - K^*)} - \frac{k_{12} S^* K^*}{K_{12} + K^*} - k_d K^*$$

$$\dot{G}^* = \frac{k_{13} E^*(G_T - G^*)}{K_{13} + (G_T - G^*)} - k_g G^*$$

$$\dot{S}^* = \frac{k_{14} K^*(S_T - S^*)}{K_{14} + (S_T - S^*)} - k_s S^*$$

The final system of ODEs which describes this new model (Model 2) are presented as;

$$\dot{L} = -k_l L$$

$$\dot{E}^* = \frac{k_1 L(E_T - E^*)}{K_1 + (E_T - E^*)} - k_e E^*$$

$$\dot{R}^* = \frac{k_5 E^*(R_T - R^*)}{K_5 + (R_T - R^*)} - \frac{k_6 G^* R^*}{K_6 + R^*} - k_r R^*$$

$$\dot{F}^* = \frac{k_7 R^*(F_T - F^*)}{K_7 + (F_T - F^*)} - k_f F^*$$

$$\dot{M}^* = \frac{k_9 F^*(M_T - M^*)}{K_9 + (M_T - M^*)} - \frac{k_{10} S^* M^*}{K_{10} + M^*} - k_m M^*$$

$$\dot{K}^* = \frac{k_{11} M^*(K_T - K^*)}{K_{11} + (K_T - K^*)} - \frac{k_{12} S^* K^*}{K_{12} + K^*} - k_d K^*$$

$$\dot{G}^* = \frac{k_{13} E^*(G_T - G^*)}{K_{13} + (G_T - G^*)} - k_g G^*$$

$$\dot{S}^* = \frac{k_{14} K^*(S_T - S^*)}{K_{14} + (S_T - S^*)} - k_s S^*$$

Linearization of the system ODEs for Model 2

The linearized system is given by;

$$\dot{\vec{X}} = J_X \vec{X}$$

This system has only one equilibrium point which is $X^* = (0, 0, 0, 0, 0, 0, 0, 0)$

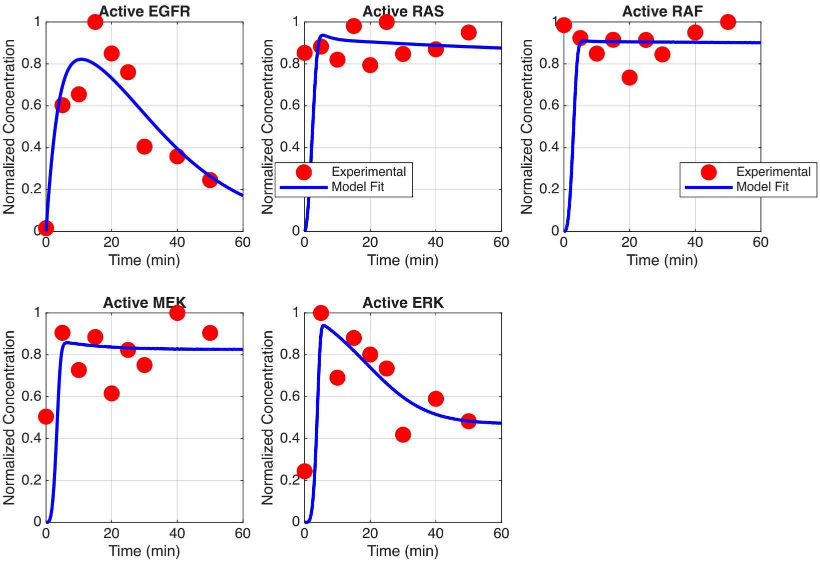
Theorem (Linear Stability of the Trivial Equilibrium)

The unique equilibrium point $X^* = \vec{0}$ of the linearized system is globally asymptotically stable.

The proof follows the same approach as that of the previous system.

The eigenvalues of the Jacobian matrix are all are real and negative

Validation of the System of ODEs for Model 2



From Pathways to Networks: Motivation for Graph Analysis

Rationale

Biological signaling pathways, such as the MAPK cascade, can be represented as **networks** where:

- **Nodes:** Proteins or signaling molecules
- **Edges:** Physical or regulatory interactions

Modeling the pathway as a network allows us to apply **graph theory** to systematically identify key regulators.

Key Idea

Nodes with higher *centrality* are often:

- Essential for network integrity (Jeong et al., 2001)
- Functionally conserved and regulatory (Barabási & Oltvai, 2004)
- Positioned to control information flow (Koschützki & Schreiber, 2008)

→ Using centrality-based ranking, we can prioritize proteins as potential drug targets.

Graph Centrality Measures and Biological Interpretation

Measure	Mathematical Definition	Biological Relevance / Interpretation
Degree	$C_D(v) = \deg(v)$	Number of direct interactions; high-degree nodes (hubs) are often essential proteins or multifunctional regulators (Jeong et al., 2001).
Betweenness	$C_B(v) = \sum_{s \neq v \neq t} \frac{\sigma_{st}(v)}{\sigma_{st}}$	Fraction of shortest paths passing through v; identifies bottleneck or control nodes mediating signal flow (Koschützki & Schreiber, 2008).
Closeness	$C_C(v) = \frac{1}{\sum_t d(v,t)}$	Measures proximity to all other nodes; reflects how rapidly a protein can influence the network (Ashtiani et al., 2018).

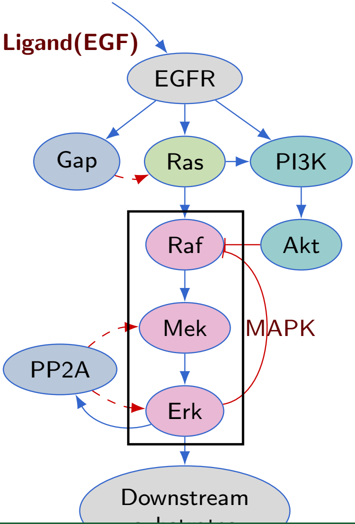
Graph Centrality Measures and Biological Interpretation

PageRank	$PR(v) = \alpha \sum_{u \in N(v)} \frac{PR(u)}{\deg(u)} + (1-\alpha)$	Weighted measure of global influence; identifies key signaling mediators or feedback nodes in pathways (Ashtiani et al., 2018).
Hub Score	From SVD of adjacency matrix: $A = HA^T$, $H = AA^T$	Nodes pointing to authoritative proteins; correspond to regulators influencing key effectors.
Authority Score	$A = A^T H$, $A = A^T A$	Nodes receiving many links from hubs; represent core signaling effectors or downstream targets. (Barabási & Oltvai, 2004).

References: Jeong et al., *Nature*, 2001; Barabási & Oltvai, *Nat Rev Genet*, 2004; Koschützki & Schreiber, *Gene Regul Syst Biol*, 2008; Ashtiani et al., *BMC Syst Biol*, 2018.

Graph Theory Analysis of Signaling Networks

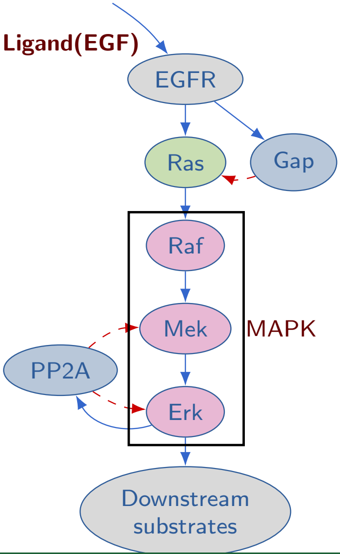
Graph Analysis on Model 1



Top Targets:

- Primary: R* (Betweenness: 10.0, Hub: 0.285)
- Secondary: F* (Highest Betweenness: 15.0)

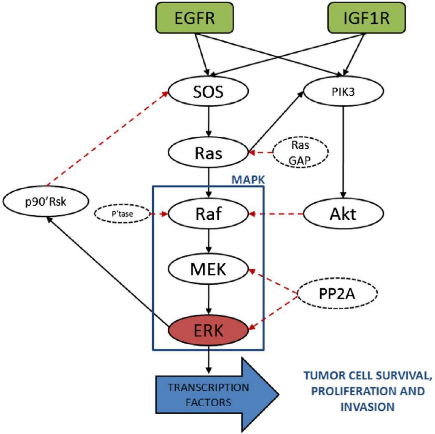
Graph Analysis on Model 2



Top Targets:

- Primary: M* (Hub: 0.250, Authority: 0.499)
- Secondary: S* (Highest Hub: 0.500)

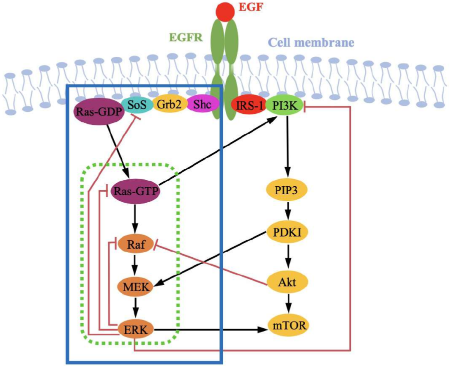
Graph Analysis on MAPK in Bianconi et al (2012)



Top Targets:

- Primary: SOS (Highest Betweenness: 26.0)
- Secondary: P* (PI3K) (Highest Authority: 0.512)

Graph Analysis on MAPK in Feng et al (2024)



Top Targets:

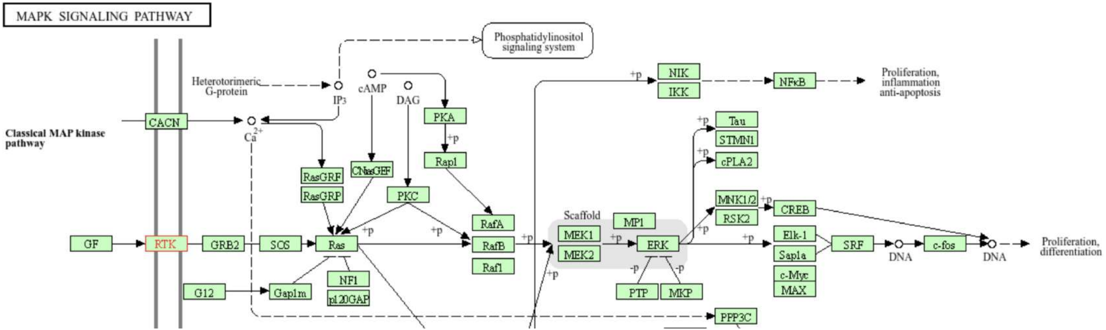
Primary: R* (Betweenness: 36.0, Hub: 0.211)

Secondary: K* (Master regulator, Out-Degree: 5)

Graph Analysis on MAPK from STRINGS

Top Targets:

- Primary: R* (Authority: 0.563, Betweenness: 80.0)
- Secondary: F* (Highest overall Betweenness: 91.0)



Conclusions and Future Directions

Conclusion

Our findings show that it is possible to construct a low-dimensional model capable of explaining the complex network dynamics of the Mitogen-Activated Protein Kinase (MAPK) pathway. This is significant because dysregulation of this pathway causes uncontrolled cell division, leading to cancer progression.

Future Directions

- We aim to further investigate the optimal approach for incorporating the ligand (L).
- Graph-based measures will be applied to the network nodes in order to rank them according to their relative importance as potential drug targets.

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Thank you for your Attention

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QUESTIONS ??